

GOVERN.AI

The Sovereign Control Plane for Enterprise AI

AUDIT TECNICO

Versione: MVP 3.0

Data: 14 Maggio 2026

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1. Stack Tecnologico

Componente	Tecnologia	Versione
Backend	FastAPI	0.110.1
Frontend	React + Tailwind	19.0.0
Database	MongoDB (Motor async)	7.0
LLM	litellm (configurable)	1.80.0
Auth	JWT HS256 + bcrypt	—
Export	ReportLab	4.1.0
Charts	Recharts	3.6.0
CI/CD	GitHub Actions	4 jobs

2. Architettura

- Backend modulare: server.py + 10 file route + models.py + database.py + seed.py + exporters.py
- Service layer disaccoppiato: services/compliance_engine.py (Compliance Intelligence Engine)
- Frontend SPA: 12 pagine, componente CRUD generico, 140+ chiavi i18n EN/IT
- Database: 7 collections, 15+ indici ottimizzati
- Autenticazione JWT con RBAC (4 ruoli: admin > dpo > auditor > viewer)
- Rate limiting su tutti gli endpoint (SlowAPI)
- Security headers middleware (5 header)

3. Collections MongoDB

Collection	Documenti	Indici
users	1+	id (unique), username (unique)
agents	14	id (unique), status, risk_level
policies	20+	id (unique), regulation, severity
audit_logs	150+	id (unique), timestamp, agent_name, outcome, risk_level
compliance_standards	8	id (unique), code (unique)
chat_messages	var	session_id + timestamp (compound)
sox_controls	20	id (unique), domain, status

4. Sicurezza

Meccanismo	Stato
JWT Auth (HS256, 8h)	IMPLEMENTATO
RBAC 4 ruoli	IMPLEMENTATO
Rate Limiting (SlowAPI)	IMPLEMENTATO
CORS restrittivo	IMPLEMENTATO
Security Headers (5)	IMPLEMENTATO
Sanitizzazione regex	IMPLEMENTATO
Enum Pydantic (10)	IMPLEMENTATO
LLM error masking	IMPLEMENTATO

5. Test & Qualita

50/50 test backend passanti (pytest) — Copertura: Auth, CRUD, Audit, Compliance, SOX Wizard, Policy Engine, Policy Guidance, Compliance Intelligence Engine, Dashboard, Chat

CI/CD: GitHub Actions con 4 job paralleli (backend-tests, frontend-build, security-scan, docker-build)

9 iterazioni testing agent, tutte passate con 100% success rate

6. Feature v3.0

- **SOX Foundation (v2.1):** Standard SOX, agente SOX Auditor, 3 policy SOX, cluster audit
- **SOX 404 Wizard (v2.2):** 20 controlli in 5 domini, progress tracking, export PDF, edit dialog
- **D.Lgs. 262 + Readiness Score (v2.3):** 8o standard, 2 policy, Dirigente Preposto, score pesato
- **Policy Conflict Engine (v2.4):** 4 regole detection, 3 endpoint, UI completa, resolve dialog
- **Policy Guidance Engine (v2.5 / Step E5):** Guidance contestuale, impact analysis, note di risoluzione obbligatorie, audit log integrato
- **Compliance Intelligence Engine (v3.0 / Step E6):** Scoring deterministico (0-100), explainability layer, score history, drivers positivi/negativi
- **Intelligence Center (v3.0):** Dashboard premium investor-ready con score ring, ranking agenti, score standard, insights e remediation

7. Debito Tecnico Residuo

ID	Area	Problema	Priorita
TD19	Database	Date come stringhe ISO	P2
TD-FE1	Frontend	Nessun test unitario frontend	P2
TD-BE1	Backend	Query dashboard non aggregate	P2
TD-BE2	Backend	Paginazione audit solo backend	P2